Functional Requirements:

- The system must have role-based access.
- Customers must be able to open new accounts with the bank .
- Customers must be able to withdraw money, deposit money, transfer money, check account balance and view all the transaction history of each of their accounts.
- The system should only allow organisations/companies to have a Saving Account and Investment Account, while individuals can have both a Savings Account and Investment Account, as well as a Cheque Account if they are working.
- The system should require an initial deposit of P500 to open an Investment Account only.
- The system should only allow a withdraw once every 2 years from a Savings Account, and once every 6 months from an Investment Account.
- An individual creating an account for the first time should provide their ID details(Names, Age, Gender, ID Number, Place of Birth, Nationality, Colour of Eyes), contact details(phone numbers, postal address, home address), next of kin details, source of income and information about place of work(where you work, salary amount, how long you have been employed there) as well as any other sources of income if any.
- A company/organization creating an account for the first time should provide their company business registration(Everything registered at CIPS, business no, etc.), information of the shareholders(everything required of the individuals but for each shareholder now) as well as company profile and turnover.
- A cheque account must have a charge of P1 for every P100 and a charge of P2 for every deposit, and a monthly transaction limit od BWP10000.
- The system should calculate interest amount daily for Savings Accounts and Investment Accounts and credit it at the end of the month(after 30 days).
- Savings Account monthly interest is 0.05%, Investment Account interest is 5%.
- The system should allow an account to be closed.

Interview with AppleBank™

1) Should a customer be able to open multiple of the same accounts?
   For simplicity, no.

2) Should transaction history be kept for the customer to view?
   Yes, all the history from when the account is made till date.

3) Should a customer be able to check their account's balance?

Yes

4) Apart from personal information, is there any additional information the bank would like to get from the customer?
   Any other additional sources of income.

5) What information does the bank collect from individual customers when opening an account?
   Names, everything on their identity cards,

6) What information does the bank collect from organisations/companies when opening accounts?
   Company Business Registration details, information of all the shareholders, the company's profile and yearly turnover.

7) Should a customer have to provide all their information for each account they open?
   Once they register once, the existing information can be reused.

8) Are there any other roles your bank tellers should be able to carry out apart from creating accounts for customers?
   View balance, deposit, withdraw, transfer money, and generate account transaction history on behalf of the customer.

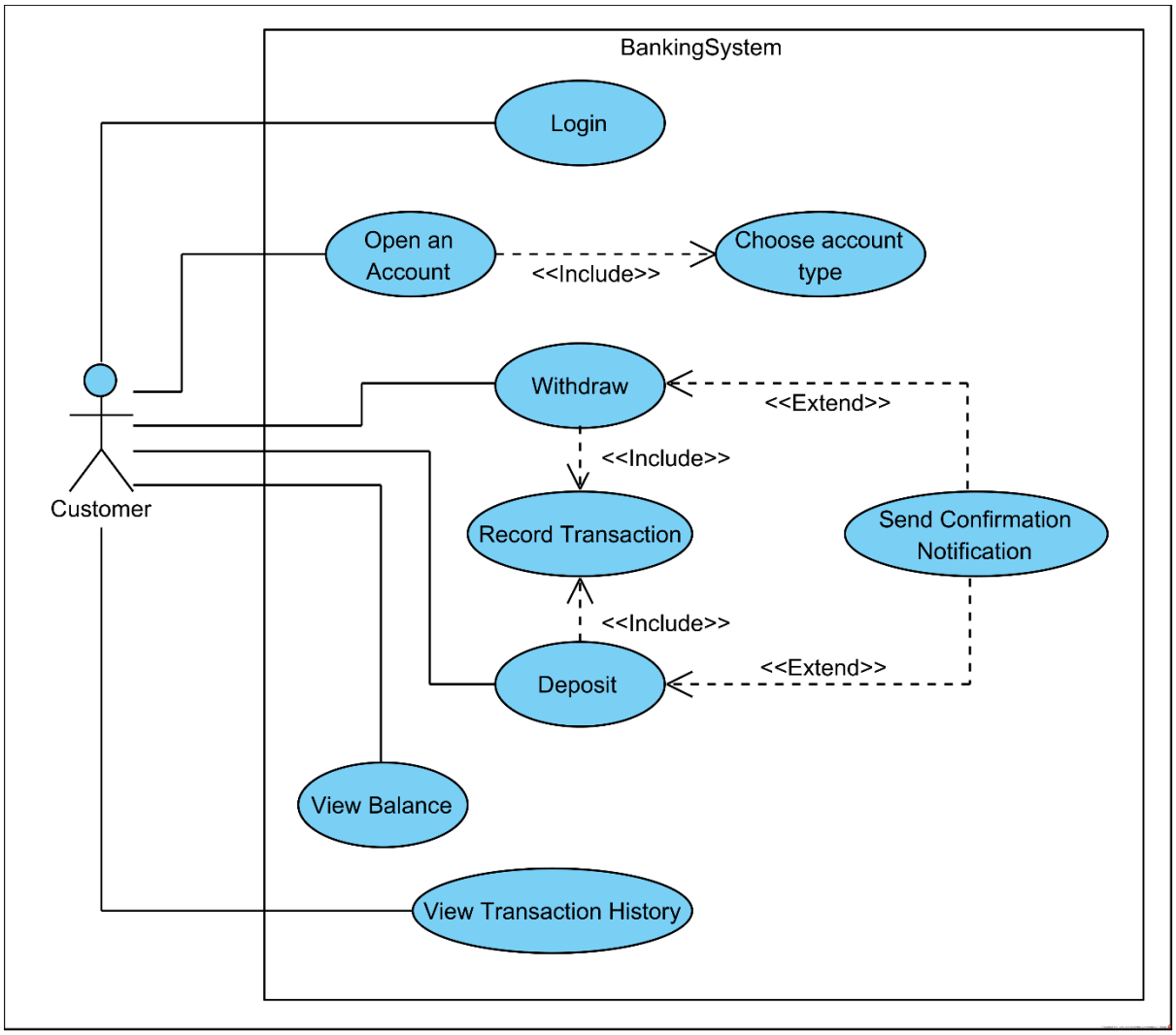
9) Does the bank have any withdraw and deposit limits?
   10000bwp a month.

10) Does the bank have any service charges for deposits and withdrawals?
    P1 for every P100 withdrawn, P2 for every deposit.

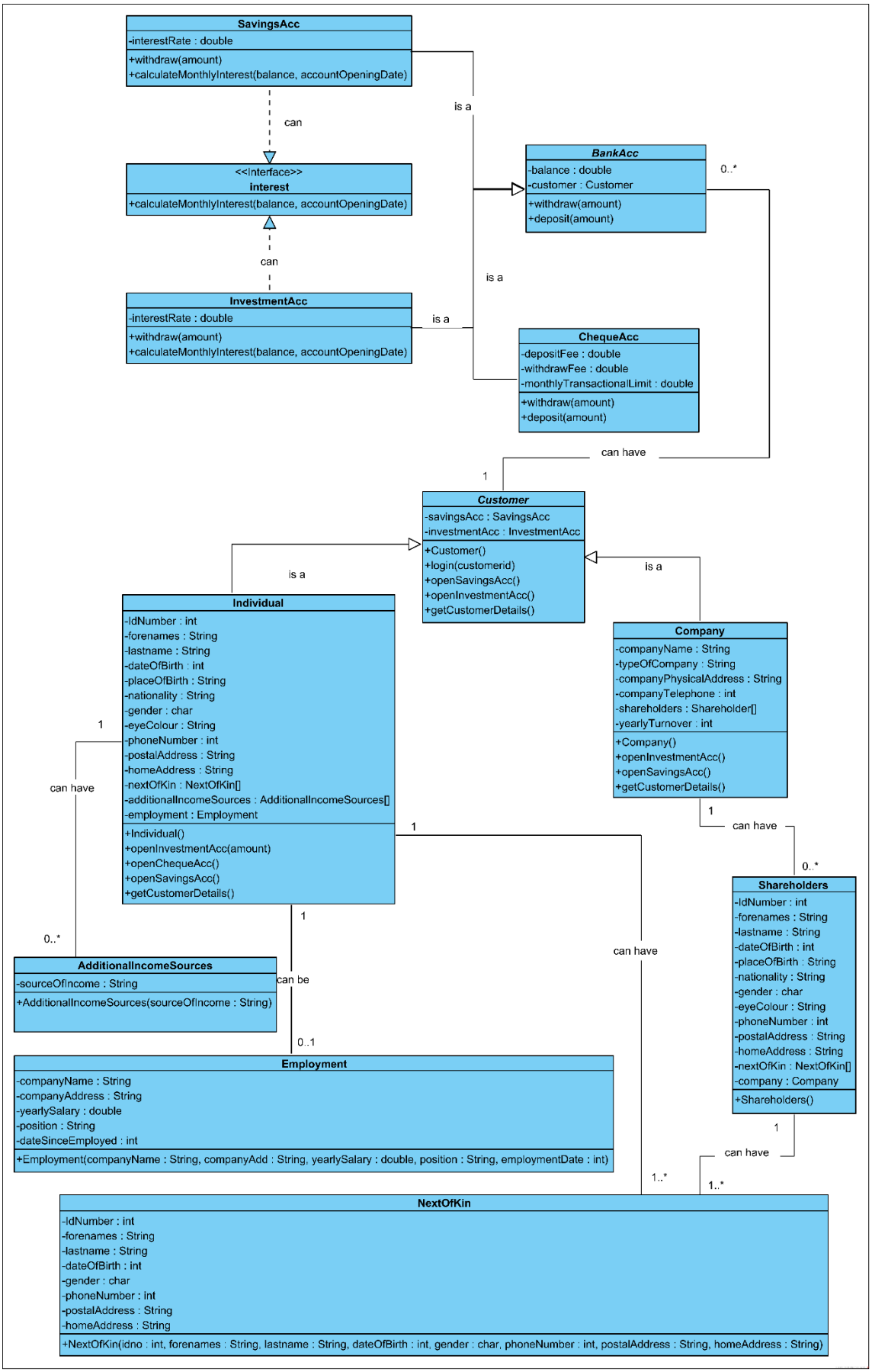
Non-Functional Requirements:

- The system should be secure.
- The system should be quick and responsive.
- The system should provide accurate data.
- The system should be optimized for smartphones, tablets, laptops and desktop computers.
- The system should have little to no downtime, if parts of the system does break, you should be able to turn off just those parts and keep the rest of it up.
- The interface should not be too complicated to reduce learning curve.

System Use Case Diagram



Class Diagram



Sequence Diagram

Diagram 1 modelling an account being opened:

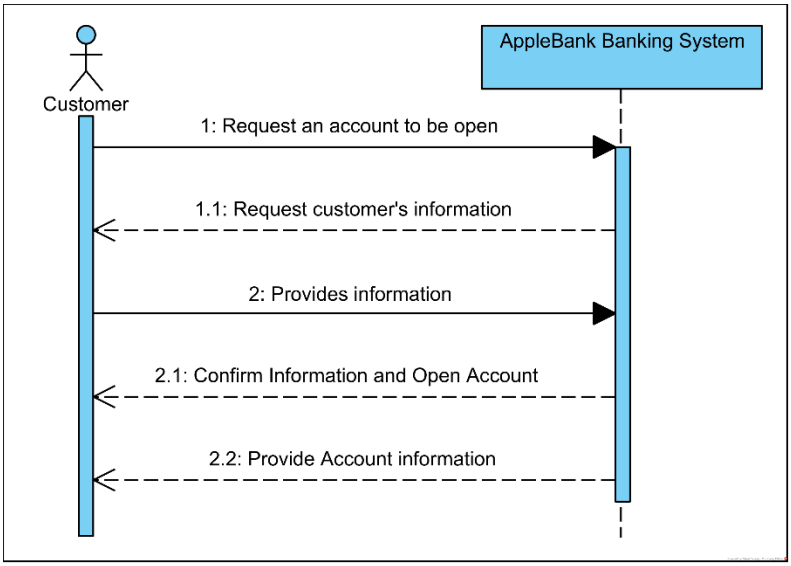
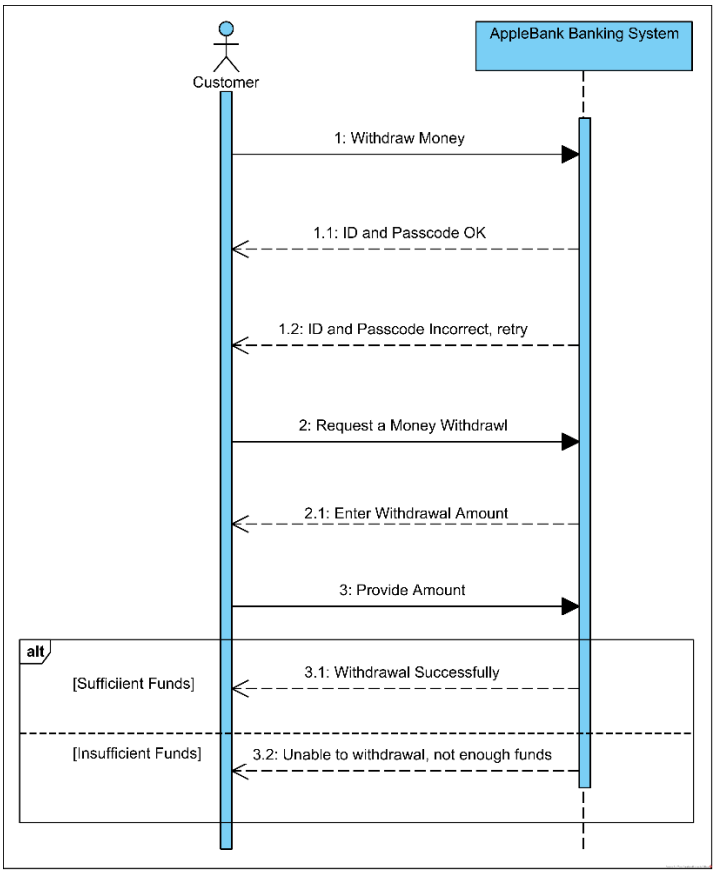


Diagram 2 modelling a transfer from one cheque account to another:



State Diagram

